

PROF. DR.-ING. MARK SCHUTERA

NUMERICAL METHODS

UNFINISHED LECTURE NOTES

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Global Optimization

2026-05-06 · cheerful mango Haubentaucher

AND THEN THERE WERE MANY.

The Why

IN THE PREVIOUS CHAPTER, we developed Newton's method and gradient descent for finding roots. The methods we learned about are local methods: They converge to whatever solution is nearby. But what if the nearby solution is bad? Or if another solution is much better but farther away?

TIME TO DO AWAY WITH OVERSIMPLIFICATIONS. In this chapter, we will:

- Learn about Shekel's foxholes, a classic test function for global optimization
- Shortly show, how we can formulate an optimization problem as a root-finding problem.
- Introduce strategies for finding global minima

IN MACHINE LEARNING AND ESPECIALLY DEEP LEARNING, understanding multi-dimensional and global optimization is crucial for training large models effectively, and to make them converge, and actually learn something useful. Neural network loss landscapes¹ are highly non-convex, do have flat regions, and contain many local minima, making optimization challenging.

VISUALIZATION OF NEURAL NETWORK LOSS LANDSCAPES Li et al. (2018) shows the complex terrain of neural network optimization. Different initializations lead to different final solutions.

¹ H. Li, Z. Xu, G. Taylor, C. Studer, and T. Goldstein. Visualizing the loss landscape of neural nets. 31:6389–6399, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/a41b3bb3e6b050b6c9067c67f663b915-Paper.pdf

Hands On Experience

SHEKEL'S FOXHOLES is a classic test function for global optimization with controlled multimodality. In 1D, it takes the simple form:

$$f(x) = - \sum_{i=1}^m \frac{c_i}{(x - a_i)^2 + r_i} \quad (1)$$

where a_i are the "foxhole" locations, c_i controls the depth of each hole, and r_i controls the width. By adjusting these parameters, we can create a landscape with multiple local minima and one global minimum, making it an ideal playground for testing optimization algorithms. Consider a specific example with three foxholes:

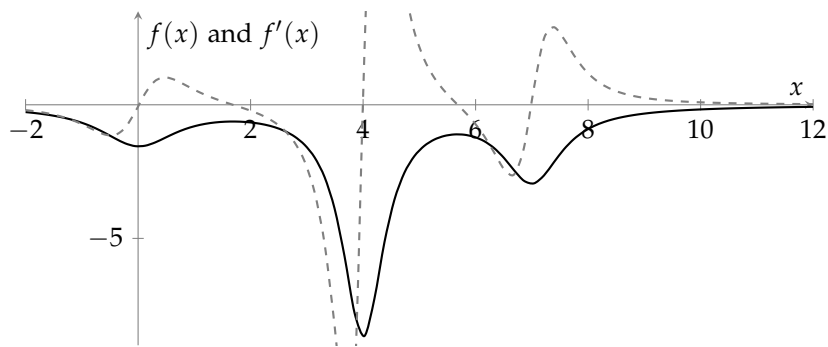


Figure 1: A 1D Shekel's foxholes function (black) with three local minima and its derivative (gray dashed). The function has valleys at $x = 0, 4, 7$, with the deepest (global minimum) at $x = 4$. The derivative (dashed line) crosses zero at each minimum.

$$f(x) = -\frac{1}{(x-0)^2 + 0.7} - \frac{1.7}{(x-4)^2 + 0.2} - \frac{1.1}{(x-7)^2 + 0.4}$$

THE GLOBAL MINIMUM is at $x = 4$. The other two minima at $x = 0$ and $x = 7$ are local minima, respectively. This information is of course not available to us when we run an optimization algorithm. We only see the function values and gradients at the points we evaluate. But the knowledge comes in handy for exploring the behavior of optimization methods.

THE SENSITIVITY TO INITIAL CONDITIONS of x_0 is something you already experienced. If we start Newton's method near $x = 0$ or $x = 7$, we'll converge to a local minimum, not the global one at $x = 4$.

LET'S APPLY NEWTON'S METHOD to our 1D Shekel function² from different starting points and observe where the method converges. Let's show it for one starting point, try another one yourself, and then we will discuss the general behavior.

THE DERIVATIVE, illustrated as the dashed line is the answer to formulating optimization as root finding. The minima of $f(x)$ correspond to the roots of $f'(x)$.

² J. Shekel. Test functions for multimodal search techniques. 1971

FROM $x_0 = 7.2$: Newton's method converges to the local minimum at $x \approx 7$. To apply Newton's method to optimization, we solve the root finding problem on $f'(x) = 0$, to reformulate the optimization problem. The Newton update becomes:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)} \quad (2)$$

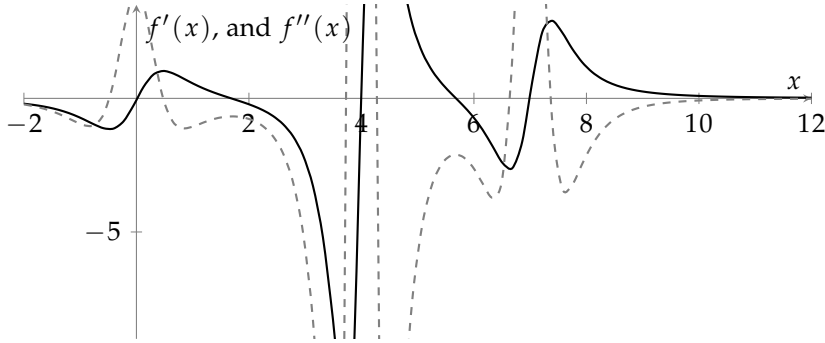


Figure 2: A 1D Shekel's foxholes function's first derivative (black), and second derivative (gray dashed). The function has valleys at $x = 0, 4, 7$, with the deepest (global minimum) at $x = 4$. The first derivative crosses zero at each minimum, and the second derivative is positive at minima (confirming local curvature).

The derivatives simplify to:

$$f'(x) = \frac{2x}{(x^2 + 0.7)^2} + \frac{3.4(x - 4)}{((x - 4)^2 + 0.2)^2} + \frac{2.2(x - 7)}{((x - 7)^2 + 0.4)^2}$$

$$\begin{aligned} f'(7.2) &\approx 0.005 + 0.100 + 2.27 \\ &= 2.38 \end{aligned}$$

$$f''(x) = \frac{2(0.7 - 3x^2)}{(x^2 + 0.7)^3} + \frac{3.4(0.2 - 3(x - 4)^2)}{((x - 4)^2 + 0.2)^3} + \frac{2.2(0.4 - 3(x - 7)^2)}{((x - 7)^2 + 0.4)^3}$$

$$\begin{aligned} f''(7.2) &\approx -0.002 - 0.091 + 7.23 \\ &= 7.14 \end{aligned}$$

$$\begin{aligned} x_1 &= 7.2 - \frac{2.38}{7.14} \\ &\approx 6.87 \end{aligned}$$

$$x_2 \approx 7.02$$

$$x_3 \approx 7.00$$

See how root finding methods can be used for optimization by applying them to the derivative of the function. The derivative $f'(x)$ is zero at extrema, so at this point we find minima or maxima.

LOCAL MINIMA NOTATION: The asterisk (*) is conventionally reserved for global optima. To distinguish local from global minima, consider alternative notation like x° (x-circle) or x_i for the i -th local minimum.

Here, we were unlucky, and the method is attracted to the nearby foxhole at $x^\circ \approx 7$.

THE FUNDAMENTAL PROBLEM HOWEVER IS that local optimization only guarantees finding a nearby extrema. Even with convergence guarantees, we would have got lucky starting from $x_0 = 3$, but from $x_0 = -2$ or $x_0 = 8$, we miss the global minima. Enough motivation for the global optimization strategies we'll discuss next.

THE LEARNING OBJECTIVES of this chapter aim at providing you with the abilities to:

- Explain why local methods cannot guarantee finding global minima
- Apply strategies for finding global extrema
- Understand how gradient descent helps to direct optimization towards minima

Global Optimization Strategies

Let us formalize local and global extrema.

LOCAL OPTIMIZATION finds a nearby minimum:

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathcal{N}(\mathbf{x}_0)} f(\mathbf{x}) \quad (3)$$

where $\mathcal{N}(\mathbf{x}_0)$ is some neighborhood of the starting point.

GLOBAL OPTIMIZATION finds the best minimum:

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \Omega} f(\mathbf{x}) \quad (4)$$

where Ω is the entire search domain. It is easy to see how the search domain can be infinitely large.

CONVEXITY IS THE DIVIDING LINE. For convex problems, any local method will find the global optimum. The difficulty arises in non-convex optimization, where the landscape contains multiple local minima, saddle points, and plateaus.

RANDOM RESTARTS is the simplest global strategy, in its notion it feels like brute-forcing again:

Require: Search domain Ω , local optimizer LocalOptimize, restarts K

```

1:  $\mathbf{x}_{\text{best}} \leftarrow \text{None}; f_{\text{best}} \leftarrow +\infty$ 
2: for  $k = 1$  to  $K$  do
3:    $\mathbf{x}_0 \leftarrow \text{RandomPoint}(\Omega)$ 
4:    $\mathbf{x}^* \leftarrow \text{LocalOptimize}(\mathbf{x}_0)$ 
5:   if  $f(\mathbf{x}^*) < f_{\text{best}}$  then
6:      $\mathbf{x}_{\text{best}} \leftarrow \mathbf{x}^*; f_{\text{best}} \leftarrow f(\mathbf{x}^*)$ 
7:   end if
8: end for
9: return  $\mathbf{x}_{\text{best}}$ 
```

If the basin of attraction of the global minimum has probability p , then with K restarts:

$$P(\text{find global}) = 1 - (1 - p)^K \quad (5)$$

For $p = 0.1$ and $K = 20$: $P = 1 - 0.9^{20} \approx 0.88$. Usually it is not possible to know p in advance. A heuristic to set K without the knowledge of p is to increase K until the best solution stabilizes (no improvement after several restarts). Patience, is a common hyperparameter for this heuristic, which controls how many restarts we wait without improvement before stopping.

$\mathcal{N}$ IS A SET of points around $\mathbf{x}_0$. For example, $\mathcal{N}(\mathbf{x}_0) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| < \epsilon\}$ for some radius ϵ .

CONVEX VS. NON-CONVEX: A function f is *convex* if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in [0, 1]$. For convex functions, every local minimum is the global minimum—local methods suffice. Non-convex functions (like Shekel’s foxholes or neural network loss surfaces) are the hard case.

Algorithm 1: Random Restarts

NEURAL NETWORK INITIALIZATION and training multiple networks with different seeds is implicit random restarts. However, this is not practical for large models.

BASIN HOPPING explores the landscape by alternating between local optimization and random jumps. This is different from random restarts, which completely resets the search. Basin hopping allows us to escape local minima while still leveraging local optimization to find better solutions.

Require: Initial point $\mathbf{x}_0$, local optimizer LocalOptimize, jump distribution for δ , iterations K

```

1:  $\mathbf{x}_{\text{best}} \leftarrow \mathbf{x}_0; f_{\text{best}} \leftarrow f(\mathbf{x}_0)$ 
2:  $\mathbf{x}_{\text{current}} \leftarrow \mathbf{x}_0$ 
3: for  $k = 1$  to  $K$  do
4:    $\mathbf{x}^* \leftarrow \text{LocalOptimize}(\mathbf{x}_{\text{current}})$ 
5:   if  $f(\mathbf{x}^*) < f_{\text{best}}$  then
6:      $\mathbf{x}_{\text{best}} \leftarrow \mathbf{x}^*; f_{\text{best}} \leftarrow f(\mathbf{x}^*)$ 
7:   end if
8:   Sample jump:  $\delta \sim \mathcal{D}$ 
9:    $\mathbf{x}_{\text{current}} \leftarrow \mathbf{x}^* + \delta$  ▷ Random jump
10: end for
11: return  $\mathbf{x}_{\text{best}}$ 

```

Algorithm 2: Basin Hopping

A jump distribution is specified as $\delta \sim \mathcal{D}$, which could be a Gaussian distribution centered at zero, or a uniform distribution over a certain range.

SIMULATED ANNEALING combines basin hopping with a probabilistic acceptance criterion for accepting point candidates. In the beginning, when hopping into a worse solution, it is accepted with probability $\exp^{-\Delta f/T}$, where T (temperature) decreases over time. Later on, the algorithm becomes more conservative and only accepts candidate points that improve the objective, allowing it to converge to an optimum.

STOCHASTIC METHODS add noise by approximating the loss landscape to escape local optima. Let's approximate the 1D shekels function as a second order function by picking three points on the foxhole curve.

THE TAKE AWAY is that the loss landscape is not static, but is to be engineered and formed. By approximating the loss landscape on mini-batches (different selection of points), we get a noisy estimate of the true loss landscape, which can help us escape local minima. The loss landscape is then of course heavily influenced by the choice

NOISY NEWTON follows the same idea around hopping, by adding Gaussian noise to the Newton step: $\mathbf{x}_{n+1} = \dots + \xi_n$.

Δf , is the increase in the objective function value when moving from the current solution to a worse candidate solution.

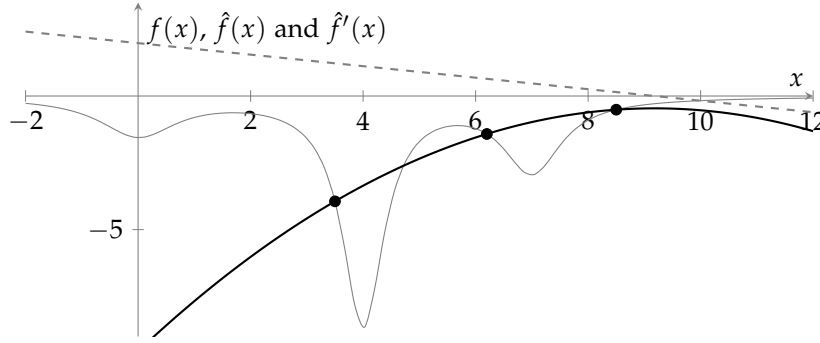


Figure 3: The original Shekel function (gray thin) with a second-order approximation (solid black) fitted through three sample points (dots). The derivative of the approximation is shown as a gray dashed line. The quadratic captures the general trend but smooths out the individual foxholes.

of the loss function itself. That being said, the Newton step is very sensitive to noise, we will see other approaches that deal better with this.

Newton's Method for finding Minima

IN THE HANDS-ON, we saw that reformulating optimization as root finding allows us to apply Newton's method to find optima:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}. \quad (6)$$

MINIMA, MAXIMA, AND SADDLE POINTS. Newton's method finds points where $f''(x) = 0$, but these can be minima, maxima, or saddle points. The second derivative test distinguishes them:

- (a) $f''(x^*) > 0$: local minimum (curve bends upward)
- (b) $f''(x^*) < 0$: local maximum (curve bends downward)
- (c) $f''(x^*) = 0$: inconclusive (possible saddle point or inflection point)

In order to direct the Newton method towards minima only, we can modify the update as such:

$$x_{n+1} = x_n - \frac{f'(x_n)}{|f''(x_n)|}. \quad (7)$$

In this way we ensure that we always move in the direction of decreasing $f(x)$. This forces the step to always move in the direction of the negative gradient (downhill so to speak).

IN DEEP LEARNING, we typically use local methods (SGD, Adam) but hope that: (1) most local minima are roughly equally good, (2) overparameterization makes bad minima rare, (3) mini-batch noise helps escape sharp minima, (4) adaptive learning rates help navigate plateaus.

How WOULD YOU modify the update to find maxima instead?

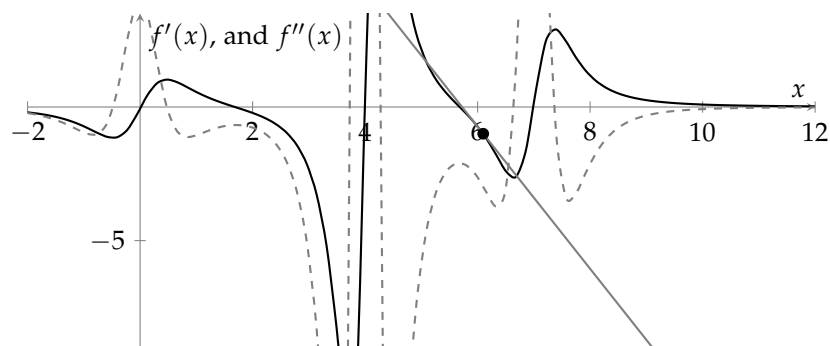


Figure 4: Solid black: the first derivative $f'(x)$ of the 1D Shekel function; gray dashed: the second derivative $f''(x)$. The black marker at $x = 6.1$ is a sample location; the solid gray line is the tangent to $f'(x)$ at that point (slope ≈ -2.66). The sign and magnitude of $f''(x)$ determine the Newton update direction and step size when solving $f'(x) = 0$.

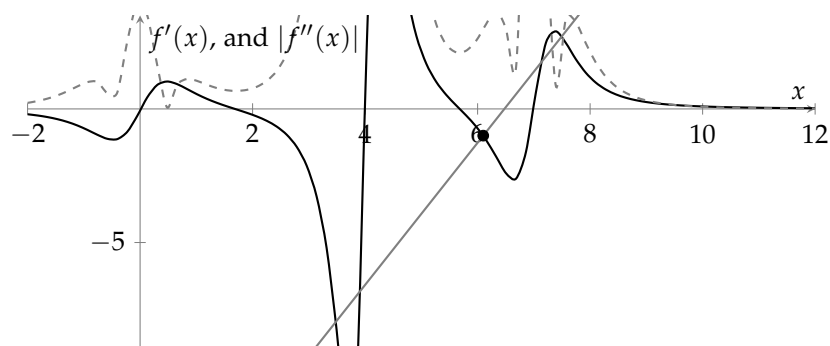


Figure 5: Solid black: the first derivative $f'(x)$; gray dashed: the absolute curvature $|f''(x)|$. Marker at $x = 6.1$ shows the same sample point as above; the solid gray line illustrates replacing $f''(x)$ by $|f''(x)|$ (slope $\approx +2.66$). Using $|f''(x)|$ in the Newton denominator removes the curvature sign, enforcing downhill steps and reducing the chance of stepping toward a local maximum.

Examples & Exercises

LET'S PUT THE STRATEGIES TO WORK on concrete numbers. Suppose you are optimizing a function with 5 local minima, and the basin of attraction of the global minimum covers 15% of the search domain ($p = 0.15$).

What is the probability of finding the global minimum with $K = 10$ random restarts?

CODE can be found at <https://github.com/Quillstacks/lecturecode-numericalmethods.git>.

HINT: Use $P = 1 - (1 - p)^K$ and solve for K : $K = \frac{\ln(1-P)}{\ln(1-p)}$.

$$\begin{aligned} P &= 1 - (1 - 0.15)^{10} = 1 - 0.85^{10} \\ &= 1 - 0.1969 = 0.803 \end{aligned}$$

About 80% chance, not too bad, but far from certain. How many restarts do we need for 99%?

$$\begin{aligned} K &= \frac{\ln(1 - 0.99)}{\ln(1 - 0.15)} = \frac{\ln(0.01)}{\ln(0.85)} \\ &= \frac{-4.605}{-0.1625} \approx 28.3 \end{aligned}$$

So $K = 29$ restarts. Now compute K for $p = 0.05$ and $p = 0.01$ at the same 99% target.

ON YOUR MACHINE, you will first design your own custom 1D Shekel function. Keep it simple in the beginning, but come back later to make it more complex. Then first familiarize yourself with the local optimization methods you have learned so far, and apply them to your function. Show where Newton's method fails (again). Then explore different x_0 and see how the convergence behavior changes. Think again about the loss landscape and basin boundaries.

DIRECTING THE SEARCH TOWARDS MINIMA. Remember the $|f''|$ trick, flips the sign of the curvature in the denominator, forcing the step to always go downhill. Think about what that means geometrically for the tangent line construction. Then try it in code.

ESCAPING LOCAL MINIMA, by global optimization strategies. Deploy random restarts and basin hopping on your custom 1D Shekel function, and compare how many function iterations each method needs to find the global minimum. How prone are they to getting stuck in local minima? How does the choice of the step size distribution affect basin hopping's performance?

NOISE AND LANDSCAPE ENGINEERING. The loss landscape is not static, but is to be engineered and formed. By approximating the loss landscape on mini-batches (different selection of points),

CODE EXERCISE: implement random restarts and basin hopping on the 1D Shekel function. Compare how many function evaluations each method needs to find the global minimum at $x \approx 4$.

FINALLY, LET'S MAP OUT THE BASINS OF ATTRACTION. Consider the 1D Shekel function and assume a local optimizer always converges to the nearest foxhole.²

- Sketch (roughly) the basins of attraction for the three foxholes at $x = 0, 4, 7$. Where are the basin boundaries?
- If basin hopping uses a jump of $\delta \sim \text{Uniform}(-3, 3)$ starting from the local minimum $x^\circ = 7$, what is the probability that a single jump lands in the basin of the global minimum?
- Why might basin hopping find the global minimum faster than random restarts here?

BASIN OF ATTRACTION. Revisit the plots of the Shekel function, mark the basins of attraction for each local minimum, and estimate their relative sizes. Which one is the global minimum? How does this relate to the probability of finding it with random restarts?

NON-CONVEX BASINS. Shekel in general is non-convex, however the basins are, note down a function, where the basins of attraction are not so easy to map out.

CAN YOU THINK OF a smart way to automatically adjust the step size distribution during optimization? What would be a good heuristic to follow? How could you prevent disastrous jumps? Implement it.

Self-Reflection and Recap

SELF-REFLECTION Questions to guide your understanding:

- Why does local optimization fail on non-convex landscapes like Shekel's foxholes?
- How does the required effort scale with the size of the global basin when doing random restarts?
- How do basin hopping differ from random restarts?
- What does $f''(x_n)$ tell us about the loss landscape, what does it characterize, and how can we put it to use?
- How does approximating the loss landscape with noisy estimates (mini-batches) help escape local optima?
- Choosing x_0 such that it lies in the basin of a local minimum, now find different ways such that your optimization method can escape it, how effective is what?

RECAP of Key Concepts:

- Local optimization methods (like Newton's method) can get stuck in local minima on non-convex functions.
- Global optimization strategies (random restarts, basin hopping, simulated annealing, stochasticity) are designed to explore the search space more broadly to find the global optimum.
- Modifying optimization methods to use curvature information $f''(x)$ allows to specify the direction of descent more accurately.
- The loss landscape is ours to engineer through the choice of loss function and noise, such as stochastic gradients.

GRADIENT DESCENT IS PRONE TO GETTING STUCK. And so far our answer was merely better than brute-forcing our way out, yet again. In the next chapter, we will learn how to smoothen loss landscape to increase the robustness of our gradient descent approach.

TEASER. What if we now face a high-frequency version of Shekel's foxholes. What is the problem that arises?